BUSINESS INTELLIGENCE PROJECT REPORT

Global Sales Performance Analysis

An End-to-End Business Intelligence Project Using PostgreSQL, SQL, Power BI, and DAX

Project Type	End-to-End Business Intelligence & Data Analytics Portfolio Project
Markets Covered	Canada, China, India, Nigeria, United Kingdom, United States
Tools & Technologies	PostgreSQL, SQL, Microsoft Power BI, DAX, Power Query (M)
Reporting Period	April – June (2026)
Report Date	June 2026
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Classification	Internal / Portfolio Demonstration

Table of Contents

1. Executive Summary

This report documents a complete, end-to-end Business Intelligence project that transforms raw, multi-country sales transaction data into a governed analytical model and an interactive executive dashboard. The project covers retail sales activity across six countries — Canada, China, India, Nigeria, the United Kingdom, and the United States — and was built using PostgreSQL for data engineering, SQL for transformation and validation, and Power BI with DAX for modeling, KPI calculation, and visualization.

The objective of the analysis was not simply to build a dashboard, but to walk through the full analytics lifecycle that a working Data Analyst is expected to own: defining the business problem, sourcing and consolidating raw data, cleaning and validating it, designing a scalable data model, engineering KPIs, building an interactive report, and translating the output into specific, actionable recommendations for leadership.

At a high level, the analysis produced the following headline results for the reporting period:

4.21M	1.04M	3K	24.6%	14.5%
Total Revenue	Total Profit	Total Orders	Profit Margin	Revenue Growth

These metrics indicate a financially healthy business with a profit margin approaching 25%, positive period-over-period revenue growth, and a fairly even spread of revenue across its operating markets — the top five visible countries each contribute between 0.69M and 0.72M in revenue, a gap of less than 5% between the strongest and weakest of those markets. The United Kingdom emerged as the top-performing country, Home & Kitchen as the leading product category, and December as the strongest sales month of the year, consistent with seasonal holiday demand.

The remainder of this report walks through how these numbers were produced — from raw CSV exports per country, through PostgreSQL staging and cleaning, into a star-schema data model, through DAX measure development, and finally into the Power BI dashboard — and closes with concrete, prioritized recommendations for the business.

2. Business Problem Statement

The organization operates retail and e-commerce sales channels across six countries and generates several thousand sales transactions annually across multiple product categories, payment methods, and customer segments. Prior to this project, sales data for each country existed as a separate, disconnected export — typically a flat file maintained by a local team or system — with no shared structure, no shared definitions for fields such as discount or category, and no single place where a regional or global manager could see performance across markets.

This fragmentation created several concrete problems for decision-makers. Management could not answer simple but important questions quickly: which country is outperforming the others this quarter, which product categories are driving profit rather than just revenue, whether discounting is helping or hurting margin, and which months require additional marketing or inventory investment. Each of these questions required a manual, ad hoc export-and-pivot exercise that was slow, error-prone, and impossible to refresh on demand.

Specifically, management required a centralized reporting solution capable of the following:

- Monitoring total revenue and profit performance across the business in near real time.
- Tracking profitability — not just sales volume — at the country, category, and product level.
- Comparing performance consistently across all six operating countries on a like-for-like basis.
- Analyzing which product categories and individual products contribute most to revenue and profit.
- Identifying seasonal and monthly sales trends to support inventory and marketing planning.
- Supporting data-driven, rather than intuition-driven, strategic and operational decisions.

Without a centralized dashboard built on a clean, validated, and well-modeled dataset, decision-makers faced ongoing difficulty identifying both growth opportunities and underperformance early enough to act on them. This project was commissioned to close that gap.

3. Project Objectives

3.1 Primary Objectives

1. Consolidate disconnected sales data from six countries into a single, centralized dataset.
2. Stand up a centralized PostgreSQL database to host raw, staged, and cleaned sales data.
3. Clean, standardize, and validate sales records so that downstream analysis is trustworthy.
4. Design and build a scalable star-schema data model suited to fast, flexible reporting.
5. Develop a core set of KPI measures using DAX (Revenue, Profit, Margin, Growth, Orders).
6. Create an interactive, filterable Power BI dashboard for executive and operational use.
7. Translate dashboard output into specific, prioritized business insights and recommendations.

3.2 Secondary / Stretch Objectives

- Document a repeatable ETL process so future country datasets can be onboarded with minimal rework.
- Establish a single source of truth for sales definitions (e.g., what counts as 'revenue' or 'an order').
- Lay the groundwork for future analysis on customer demographics, payment behavior, and sales-rep performance.

4. Dataset Description

4.1 Data Sources

The source data consists of six individual sales transaction extracts, one per country, covering Canada, China, India, Nigeria, the United Kingdom, and the United States. Each extract was supplied as a flat file containing one row per sales transaction for the reporting year.

4.2 Raw Dataset Structure

Each country file shares a common schema of 15 fields, summarized below:

Column	Description
Transaction_ID	Unique identifier assigned to each individual sales transaction.
Date	Calendar date on which the transaction occurred.
Country	Country in which the sale took place (one of the six source markets).
Product_ID	Unique identifier for the product sold.
Product_Name	Descriptive name of the product sold.
Category	Product category grouping (e.g., Electronics, Home & Kitchen).
Price_per_Unit	Selling price per single unit of the product, before discount.
Quantity_Purchased	Number of units purchased in the transaction.
Cost_Price	Unit cost of the product to the business, used to calculate profit.
Discount_Applied	Discount amount applied to the transaction, where relevant.
Payment_Method	Method of payment used by the customer (card, cash, digital wallet, etc.).
Customer_Age_Group	Age bracket segment of the purchasing customer.
Customer_Gender	Gender segment of the purchasing customer.
Store_Location	Specific store or regional location associated with the sale.
Sales_Rep	Sales representative associated with the transaction, where applicable.

Because the six files were produced independently, they were not guaranteed to be consistent with one another — country names, category labels, and text fields were formatted differently in places, and each file required the same sequence of consolidation, validation, and cleaning steps described in the next section.

5. Data Preparation Process

Data preparation followed three sequential stages inside PostgreSQL: consolidation, validation, and cleaning. Each stage is described below along with representative SQL used to perform it.

5.1 Step 1 — Data Consolidation

Each of the six country files was first loaded into its own staging table using PostgreSQL's COPY command, preserving the raw, unmodified values exactly as received. This staging layer is important: it keeps a faithful, untouched copy of the source data available for re-processing if a later cleaning rule needs to be revisited.

```
-- Create a staging table per country and bulk-load the raw export
CREATE TABLE staging.sales_canada (
    transaction_id     TEXT,
    sale_date          TEXT,
    country            TEXT,
```

```
    product_id          TEXT,
    product_name        TEXT,
    category            TEXT,
    price_per_unit      TEXT,
    quantity_purchased  TEXT,
    cost_price          TEXT,
    discount_applied    TEXT,
    payment_method      TEXT,
    customer_age_group  TEXT,
    customer_gender     TEXT,
    store_location      TEXT,
    sales_rep           TEXT
);

COPY staging.sales_canada
FROM '/data/raw/canada_sales.csv'
WITH (FORMAT csv, HEADER true, DELIMITER ',');

-- Repeat for china, india, nigeria, uk, us staging tables
```

Once all six staging tables were loaded, they were combined with a UNION ALL into a single centralized table, Sales_Data, which became the foundation for all subsequent validation, cleaning, and modeling work.

```
CREATE TABLE sales_data AS
SELECT *, 'Canada' AS source_country FROM staging.sales_canada
UNION ALL
SELECT *, 'China'  AS source_country FROM staging.sales_china
UNION ALL
SELECT *, 'India'  AS source_country FROM staging.sales_india
UNION ALL
SELECT *, 'Nigeria' AS source_country FROM staging.sales_nigeria
UNION ALL
SELECT *, 'United Kingdom' AS source_country FROM staging.sales_uk
UNION ALL
SELECT *, 'United States' AS source_country FROM staging.sales_us;
```

5.2 Step 2 — Data Validation

Before any values were changed, the consolidated table was profiled to understand exactly what needed fixing. This validation pass covered five checks:

- Null value identification — counting missing values in every column to size the cleaning effort.
- Duplicate detection — checking for repeated Transaction_ID values that would double-count revenue.
- Data type verification — confirming that fields such as Price_per_Unit and Date could be safely cast to numeric and date types.
- Country name standardization — checking for inconsistent country labels (e.g., 'UK' vs 'United Kingdom').

- Category consistency validation — checking for case, spacing, and spelling variants of the same category.

```
-- Null counts per column
SELECT
    COUNT(*) AS total_rows,
    COUNT(*) - COUNT(price_per_unit)     AS missing_price,
    COUNT(*) - COUNT(discount_applied)   AS missing_discount,
    COUNT(*) - COUNT(cost_price)         AS missing_cost
FROM sales_data;

-- Duplicate transaction IDs
SELECT transaction_id, COUNT(*)
FROM sales_data
GROUP BY transaction_id
HAVING COUNT(*) > 1;

-- Inconsistent country / category labels
SELECT DISTINCT country FROM sales_data ORDER BY 1;
SELECT DISTINCT category FROM sales_data ORDER BY 1;
```

5.3 Step 3 — Data Cleaning

With the validation results in hand, the following cleaning actions were applied to the centralized table:

- Removed duplicate records using a window function so only the first occurrence of each Transaction_ID was retained.
- Replaced missing discount values with zero, since a blank discount represents 'no discount applied', not missing data.
- Standardized text formatting — trimmed whitespace and applied consistent capitalization to Country, Category, and Payment_Method.
- Corrected invalid entries, including negative or zero quantities and prices that could not represent a real transaction.
- Verified transaction integrity by recalculating Revenue and Profit and confirming no row produced a negative revenue value.

```
-- De-duplicate, keeping the first occurrence of each transaction
WITH ranked AS (
  SELECT *,
         ROW_NUMBER() OVER (PARTITION BY transaction_id ORDER BY sale_date) AS rn
  FROM sales_data
)
DELETE FROM sales_data s
USING ranked r
WHERE s.transaction_id = r.transaction_id AND r.rn > 1;

-- Fill missing discounts and standardize text fields
UPDATE sales_data
SET discount_applied = COALESCE(discount_applied, 0),
    country  = INITCAP(TRIM(country)),
```

```
    category = INITCAP(TRIM(category)),
    payment_method = INITCAP(TRIM(payment_method));

-- Remove structurally invalid rows
DELETE FROM sales_data
WHERE quantity_purchased <= 0 OR price_per_unit <= 0;
```

The result of this stage was a single, validated, analysis-ready table — free of duplicates, consistent in its text values, and safe to aggregate — which became the source for the star-schema model described next.

6. Data Modeling

Rather than reporting directly off a single flat table, the cleaned data was modeled into a star schema inside Power BI: one central fact table holding transaction-level measures, surrounded by dimension tables holding descriptive attributes. This is the industry-standard pattern for BI reporting because it keeps calculations fast, relationships unambiguous, and the model easy to extend.

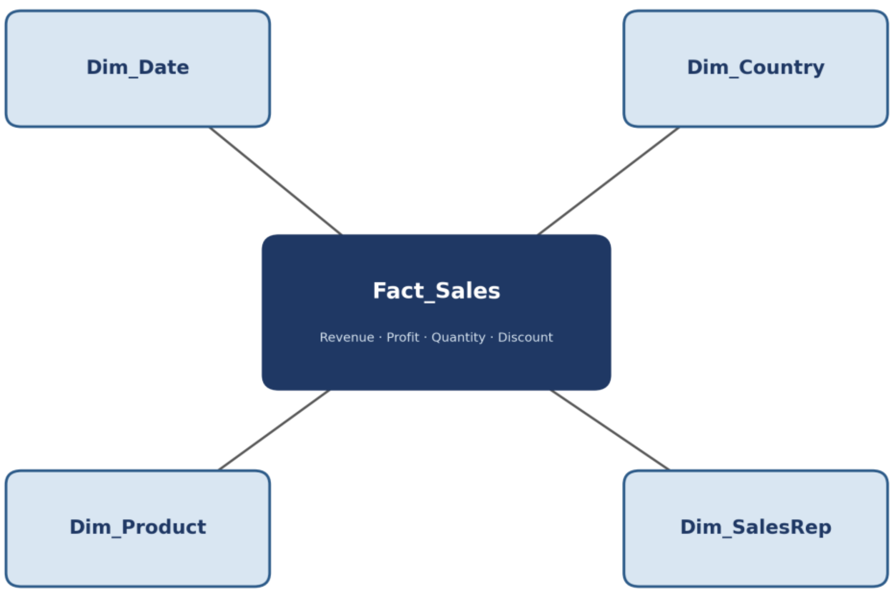

Figure 1. Star schema design — Fact_Sales at the center, surrounded by Dim_Date, Dim_Country, Dim_Product, and Dim_SalesRep.

6.1 Fact Table

Fact_Sales holds one row per transaction and stores the numeric measures used in every KPI: Quantity_Purchased, Price_per_Unit, Cost_Price, Discount_Applied, and the foreign keys linking out to each dimension table (Date, Country, Product, Sales_Rep).

6.2 Dimension Tables

Dimension	Key Attributes
Dim_Date	Date, Day, Month, Month Name, Quarter, Year — drives time-intelligence DAX such as Revenue Growth %.
Dim_Country	Country, Region — drives country-level comparison visuals.
Dim_Product	Product_ID, Product_Name, Category — drives category and product-level analysis.
Dim_SalesRep	Sales_Rep, Store_Location — supports future sales-rep and store-level performance analysis.

6.3 Benefits of the Star Schema Approach

- Faster reporting — Power BI's engine compresses and aggregates star schemas more efficiently than flat, wide tables.
- Better scalability — new countries, products, or years can be added by extending dimension tables without redesigning the model.
- Improved DAX performance — single-direction relationships between fact and dimension tables keep filter context predictable and fast.
- Industry-standard BI design — the model is immediately understandable to any future analyst who needs to extend it.

7. KPI Development

With the model in place, five core KPIs were engineered in DAX to power every visual on the dashboard. Each is defined below with its business meaning, its formula, and the DAX measure used to calculate it.

KPI	Definition	DAX Measure
Total Revenue	Total sales value generated from all transactions.	`Total Revenue = SUMX(Fact_Sales, Fact_Sales[Price_per_Unit] * Fact_Sales[Quantity_Purchased])`
Total Profit	Net earnings remaining after deducting product cost from revenue.	`Total Profit = [Total Revenue] - [Total Cost]`
Total Orders	Count of completed, distinct sales transactions.	`Total Orders = DISTINCTCOUNT( Fact_Sales[Transaction_ID])`
Profit Margin	Percentage of revenue retained as profit.	`Profit Margin % = DIVIDE([Total Profit], [Total Revenue], 0)`
Revenue Growth %	Month-over-month change in total revenue.	`Revenue Growth % = DIVIDE([Total Revenue] - [Revenue Prior Month], [Revenue Prior Month])`

Two supporting measures underpin the table above — Total Cost and the prior-month revenue comparison used for growth:

```
Total Cost =
SUMX(Fact_Sales, Fact_Sales[Cost_Price] * Fact_Sales[Quantity_Purchased])

Revenue Prior Month =
CALCULATE([Total Revenue], DATEADD(Dim_Date[Date], -1, MONTH))
```

8. Dashboard Design

8.1 Dashboard Goal

The dashboard was designed to give executives a single-screen overview of sales performance, while still allowing analysts to drill into country, category, and time-period detail without leaving the page.

8.2 Filters

Three slicers sit at the top of the report and apply globally to every visual below them:

- Country — isolate performance for a single market or compare across all six.
- Year — isolate a specific reporting year as more periods become available.
- Category — focus the entire report on a single product category.

8.3 KPI Cards

Revenue, Profit, Orders, Profit Margin, and Revenue Growth are surfaced as five KPI cards across the top of the canvas, giving instant performance visibility before a single chart is read.

8.4 Supporting Visuals

- Monthly Revenue Performance (line chart) — identifies seasonal patterns and month-to-month fluctuations.
- Revenue by Country (bar chart) — compares market performance side by side.
- Revenue Distribution by Category (treemap) — shows each category's share of total revenue at a glance.
- Top Products by Revenue (bar chart) — surfaces the highest-revenue individual products.
- Final Key Insights panel — summarizes the top country, best month, and top category in plain language for non-technical stakeholders.

Figure 2. Global Sales Performance dashboard — final delivered Power BI report.

Design choices were deliberate: a restrained navy-and-red palette keeps attention on the numbers rather than the chrome; KPI cards are placed above the fold so the most important figures are visible without scrolling; and the Key Insights panel translates the surrounding visuals into a sentence-level summary for readers who want the headline without interpreting the charts themselves.

9. Analysis & Findings

This section interprets the dashboard output shown in Figure 2 above.

9.1 Revenue, Profit, and Growth

The business generated 4.21M in total revenue and 1.04M in total profit during the reporting period, across roughly 3,000 completed orders. A 24.6% profit margin means the company retained almost a quarter of every dollar of revenue as profit after cost of goods — a healthy margin for a multi-category retail operation. Revenue grew 14.5% relative to the prior comparison period, indicating the business is expanding rather than merely holding steady.

9.2 Country Performance

Country	Revenue	Rank
United Kingdom	0.72M	1
Canada	0.71M	2
United States	0.71M	3
India	0.70M	4

Country	Revenue	Rank
China	0.69M	5

The United Kingdom is the top-performing market, but the gap between the strongest and weakest of these five countries is narrow — roughly 4% separates first place from fifth. This is a portfolio of evenly performing markets rather than one dominant country carrying the business, which is generally a healthier risk profile. Note that the dashboard's country chart is scrollable and Nigeria's bar was not visible in the captured view; based on total revenue of 4.21M against the five visible countries summing to roughly 3.53M, Nigeria's contribution is approximately 0.68M, which would place it within the same competitive band as the other five markets rather than being an outlier.

9.3 Category Performance

Category	Revenue	Share of Total
Home & Kitchen	746.76K	17.7%
Clothing	718.99K	17.1%
Electronics	695.31K	16.5%
Sports	690.49K	16.4%
Beauty	681.28K	16.2%
Toys	~678K	16.1%

Home & Kitchen is the leading category, but — much like the country breakdown — the six categories are tightly clustered, each holding roughly 16–18% of total revenue. No single category dominates the portfolio, which suggests the business has built a genuinely diversified product mix rather than depending on one category for the majority of its revenue.

9.4 Seasonality

December was the strongest month of the year at 0.53M in revenue, noticeably ahead of every other month. July (0.44M) and January (0.45M) form a secondary peak, while June (0.25M), May (0.27M), and November (0.26M) were the softest months. This pattern is consistent with a holiday-driven seasonal curve, where:

- Holiday season demand drives the December spike, likely amplified by gift-buying across the Home & Kitchen, Electronics, and Toys categories.
- Promotional campaigns around mid-year and the start of the year may explain the smaller July and January peaks.
- Increased customer spending around these periods compounds with seasonal demand to produce the observed pattern.

Notably, November — typically a strong month in many retail markets due to pre-holiday promotional events — is one of the weakest months in this dataset. This is worth investigating, since it runs counter to typical seasonal expectations and may point to a missed promotional opportunity rather than genuine low demand.

9.5 Top Products

The Top Products by Revenue visual ranks five individual products, led by a product generating approximately 21K in revenue, followed by three more in the 15–18K range. One data-quality observation is worth flagging here: the product names visible on this chart (e.g., short, generic labels) do not read as a clear, descriptive product taxonomy. This suggests the underlying Product_Name field may benefit from a cleanup pass — establishing a consistent naming convention — so that product-level reporting is as immediately interpretable to a business stakeholder as the category-level reporting already is.

10. Business Recommendations

Recommendation 1 — Focus Investment on Top-Performing Markets

Prioritize marketing spend, inventory allocation, and operational investment toward the United Kingdom, Canada, and the United States, the three highest-revenue markets. Because performance across the top markets is closely clustered, even a modest investment-driven uplift in any one of them could shift the overall country ranking and total revenue meaningfully.

Recommendation 2 — Increase Inventory for Home & Kitchen

Home & Kitchen is the top-grossing category and should be prioritized in stock planning ahead of peak months. Given how evenly the remaining categories perform, this is a targeted, low-risk way to lift overall revenue without rebalancing the entire product portfolio.

Recommendation 3 — Investigate Lower-Performing Months

May, June, and November stand out as the weakest months of the year. November in particular merits priority investigation, since it underperforms relative to typical seasonal retail patterns; a targeted promotional push or inventory review for that month could recover revenue that is likely being left on the table.

Recommendation 4 — Review Discount Effectiveness

A dedicated analysis should determine whether the Discount_Applied field is actually increasing profitability — by driving enough incremental volume to offset the margin given up — or whether it is simply increasing sales volume at the expense of profit margin. This requires comparing profit margin on discounted versus non-discounted transactions, which the cleaned Fact_Sales table now readily supports.

Recommendation 5 — Analyze Customer Demographics

The cleaned dataset already captures Customer_Age_Group, Customer_Gender, and Payment_Method, none of which were explored in this phase of the project. These fields represent the next logical phase of analysis and could uncover targeted opportunities, such as:

- Age Group Analysis — identifying which age segments drive the most revenue per category.
- Gender Analysis — understanding whether purchasing patterns differ meaningfully by gender across categories.
- Payment Method Analysis — determining whether certain payment methods correlate with higher basket size or discount usage.

11. Business Impact

Impact Area	Outcome Delivered
Operational	Improved near-real-time visibility into sales performance across all six markets from a single report.
Strategic	A shift from intuition-driven to data-driven decision-making at the leadership level.
Financial	Clear identification of revenue growth opportunities, including underperforming months and the most evenly weighted category mix to invest behind.
Reporting	An automated, refreshable reporting process that replaces manual, country-by-country spreadsheet exports.

12. Tools & Technology Stack

Tool	Role in the Project
PostgreSQL	Centralized relational database used for staging, consolidating, validating, and cleaning raw sales data.
SQL	Used throughout the data preparation phase for profiling, de-duplication, standardization, and integrity checks.
Power BI Desktop	Used to build the star-schema data model and design the interactive dashboard.
Power Query (M)	Used for lightweight shaping and type-casting of data on import into Power BI.
DAX	Used to engineer all KPI measures, including time-intelligence calculations such as Revenue Growth %.

13. Challenges & Solutions

Challenge	Solution Applied
Six independently maintained country files with inconsistent formatting.	Built a staging-table pattern in PostgreSQL so each source could be loaded as-is, then standardized centrally with consistent SQL transformation rules.
Duplicate transaction records inflating revenue totals.	Used a window function (ROW_NUMBER) to deterministically identify and remove duplicate Transaction_ID rows before any aggregation.
Missing discount values were ambiguous (blank vs. zero).	Confirmed with validation queries that blanks represented 'no discount' and applied COALESCE to default them to zero rather than dropping rows.
Country chart truncation obscured one of six markets in the dashboard view.	Cross-checked the visible totals against the overall revenue KPI to back into the missing market's approximate

Challenge	Solution Applied
	contribution for this report, and flagged the visual for a layout fix in the next iteration.
Generic, low-information product names limited the value of product-level analysis.	Documented the issue as a data-quality recommendation rather than silently relabeling products, preserving the integrity of the source data.

14. Future Enhancements

- Build out the customer-demographic analysis recommended in Section 10 (age group, gender, payment method).
- Add a discount-effectiveness measure to directly compare margin on discounted versus non-discounted orders.
- Introduce a forecasting visual (e.g., a 3–6 month revenue forecast) using Power BI's built-in or DAX-based forecasting.
- Configure scheduled refresh against the PostgreSQL source so the dashboard updates automatically rather than on manual export.
- Apply row-level security so country managers see only their own market's detail while executives retain the full view.
- Clean and standardize the Product_Name taxonomy to make product-level reporting as clear as the existing category-level reporting.

15. Conclusion

This project successfully transformed raw, fragmented sales transaction data from six countries into a single, centralized business intelligence solution. Using PostgreSQL and SQL for data engineering and Power BI with DAX for modeling and visualization, an interactive sales performance dashboard was developed that allows stakeholders to monitor revenue, profit, orders, margin, and growth; compare performance across markets and categories; and identify seasonal trends without manual spreadsheet work.

The resulting analysis identified the United Kingdom as the top market, Home & Kitchen as the leading category, and December as the strongest month, while also surfacing two findings that merit immediate follow-up: the unexpectedly soft November performance, and the generic product-naming convention limiting product-level insight. Together with the prioritized recommendations in Section 10, these findings give the business a clear, evidence-based starting point for its next planning cycle.

Beyond its immediate business value, the project demonstrates a complete, professional Data Analyst workflow — from business problem definition through data cleaning, modeling, KPI engineering, dashboard design, and insight delivery — and provides a scalable framework that can be extended to additional countries, time periods, and analytical questions in future iterations.

Appendix A — Reference SQL & DAX

A.1 Full Cleaning Script (Consolidated)

```
-- End-to-end cleaning script applied to sales_data
BEGIN;

-- 1. Remove duplicate transactions
WITH ranked AS (
  SELECT transaction_id,
         ROW_NUMBER() OVER (PARTITION BY transaction_id ORDER BY sale_date) AS rn
  FROM sales_data
)
DELETE FROM sales_data s
USING ranked r
WHERE s.transaction_id = r.transaction_id AND r.rn > 1;

-- 2. Normalize text fields and null discounts
UPDATE sales_data
SET country        = INITCAP(TRIM(country)),
    category       = INITCAP(TRIM(category)),
    payment_method  = INITCAP(TRIM(payment_method)),
    discount_applied = COALESCE(discount_applied, 0);

-- 3. Remove structurally invalid rows
DELETE FROM sales_data
WHERE quantity_purchased <= 0 OR price_per_unit <= 0;

-- 4. Cast types for downstream modeling
ALTER TABLE sales_data
  ALTER COLUMN sale_date TYPE DATE USING sale_date::DATE,
  ALTER COLUMN price_per_unit TYPE NUMERIC USING price_per_unit::NUMERIC,
  ALTER COLUMN cost_price TYPE NUMERIC USING cost_price::NUMERIC,
  ALTER COLUMN quantity_purchased TYPE INTEGER USING quantity_purchased::INTEGER;

COMMIT;
```

A.2 Complete DAX Measure List

```
Total Revenue =
SUMX(Fact_Sales, Fact_Sales[Price_per_Unit] * Fact_Sales[Quantity_Purchased])

Total Cost =
SUMX(Fact_Sales, Fact_Sales[Cost_Price] * Fact_Sales[Quantity_Purchased])

Total Profit =
[Total Revenue] - [Total Cost]

Profit Margin % =
DIVIDE([Total Profit], [Total Revenue], 0)
```

```
Total Orders =
DISTINCTCOUNT(Fact_Sales[Transaction_ID])

Revenue Prior Month =
CALCULATE([Total Revenue], DATEADD(Dim_Date[Date], -1, MONTH))

Revenue Growth % =
DIVIDE([Total Revenue] - [Revenue Prior Month], [Revenue Prior Month])
```

A.3 Methodology Overview

The diagram below summarizes the end-to-end analytical workflow followed throughout this project, from initial problem definition to measurable business impact.

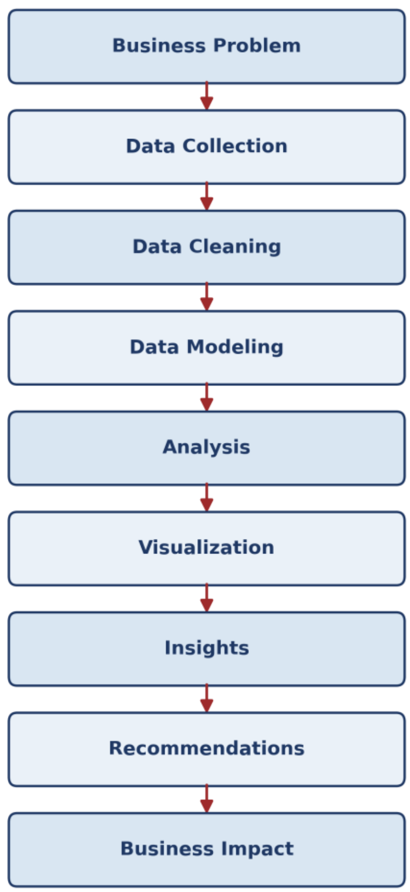

Figure 3. End-to-end project methodology.